

Submission Summary

Conference Name

2nd International Conference on Sustainability, Innovation & Technology (ICSIT 2026))

Paper ID

1044

Paper Title

SalaryLens: An Intelligent Resume-Based Salary Prediction System

Abstract

This paper presents SalaryLens, an intelligent webbased system for predicting annual salaries from resume data using advanced machine learning techniques. The system employs a stacking ensemble architecture combining XGBoost, Random Forest, Extra Trees, and Neural Networks with Ridge regression as a meta-learner. Trained on 29,850 resume records with 104 engineered features, the model achieves 97.02% R^2 score and 91.29% prediction accuracy within $\pm 10\%$ margin. The system incorporates automated resume parsing using large language models (LLMs), TF-IDF vectorization for skill extraction, and interactive what-if analysis for career planning. Experimental results demonstrate superior performance compared to individual models, with mean absolute error of \$7,071 and root mean squared error of \$8,846. The deployed web application provides real-time predictions, market comparisons, and scenario analysis, making it a practical tool for job seekers, recruiters, and HR professionals.

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Primary Subject Area

Track 2: Computational Intelligence & Machine Learning

Submission Files

SalaryLens__An_Intelligent_Resume_Based_Salary_Prediction_System.pdf (359.1 Kb, 2026-04-25 14:09:22 GMT+5:30)

Submission Questions Response

1. Originality Confirmation

Is this submission original and not under review elsewhere?

Yes

2. AI Usage Disclosure

Was AI assistance used in writing or preparing this submission?

Yes

3. Keywords

Keywords (comma-separated)

salary prediction, machine learning, stacking

ensemble, resume parsing, feature engineering, career analytics

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6. Author Consent

I confirm that all authors have approved this submission and agree to its submission

Agreement accepted

7. Format Compliance

I confirm that the submission follows IEEE conference formatting guidelines and number of page limit (6). If page limit exceeded, I am ready to pay extra page charges

Agreement accepted

8. Author Limit Policy (6 Authors)

Does this submission have 6 or fewer authors?

Agreement accepted
